🌻 Pivot Tables

Use this panel to build pivot tables and charts from your project data with a simple drag-and-drop interface. Three subtabs: **Pivot table**, **Summary** (one-dimensional heatmaps for categorical vars with <10 categories, mean/median for continuous), and **All-by-all checks** (pairwise significance tests).

Example bookmark:

- Comparing groups (gender) — heatmap pivot table

Quick start

This panel is powered by PivotTable.js, which generates most of the UI dynamically. These are the **actual widgets** you can see/use:

1. 👉 **(`Links` / `Factors` / `Sources`)** (Dropdown): `Which table to analyse?` — chooses which dataset to pivot.
2. 👉 **(`After pipeline` / `Before pipeline`)** (Radio buttons): `Which stage to analyse?` — chooses post-pipeline (matches other panels) vs raw data for the chosen dataset.
3. 👉 **(`Refresh`)** (Button): reloads + re-renders the pivot with latest data.
4. 👉 **(`Sig level`)** (Dropdown): significance threshold for 2-way table tests (0.10, 0.05, 0.01, 0.001).
5. 👉 **(`Shading`)** (Dropdown): conditional formatting for every table in the panel — see Conditional formatting.
6. 👉 **(`Copy to Clipboard`)** (Button): copies the current pivot table/chart as an image.
7. 👉 **(`Copy as Table`)** (Button): copies the pivot as tab-separated text — paste into Excel/Sheets to get a properly aligned table (**Table renderer only**).
8. 👉 **(`Download XLSX`)** (Button): downloads the current pivot as `.xlsx` (**Table renderer only**). Merged cells (rowspan/colspan) are expanded so columns align correctly.
9. 👉 **(Help)** (Button): opens help for Pivot Tables.
10. 👉 **(`Search fields`)** (Text field): searches the available field/column names in the drag-and-drop list.
11. 👉 **Drag-and-drop fields** (field chip list): shows available fields (columns). Drag a field chip into `Rows`, `Cols`, or `Vals`.
12. 👉 **(`Rows` / `Cols` / `Vals`)** (Drag-and-drop drop zones): defines how the pivot is laid out and what values are summarised.
13. 👉 **(`Aggregator`)** (Dropdown): chooses the aggregation function (e.g. `Count`, `Sum`, `Average`).
14. 👉 **(`Vals`)** (Dropdown): chooses the numeric field to aggregate (only appears for aggregators that need it).
15. 👉 **(`Renderer`)** (Dropdown): chooses the output type (table, heatmaps, Plotly charts).

16. 👉 (`Row Order` / `Col Order`) (Dropdowns): chooses how row/column keys are sorted.
17. 👉 **(Filter popup on each field chip)** (Popup): include/exclude values for that field.
    - 👉 **(Search)** (Text field): searches within the field's value list.
    - 👉 **(Checkbox list)**: ticks/unticks specific values.
18. 👉 **(× on a field chip)** (Button): removes that field from `Rows`/`Cols`/`Vals`.

Arrange fields (drag and drop)

- 👉 **(`Search fields`)**: type part of a column name to narrow the available field list.
- 👉 `Drag-and-drop list`: the "pool" of fields you can use.
- 👉 **(`Rows`)** (Drop zone): fields listed down the left side of the output.
- 👉 **(`Cols`)** (Drop zone): fields listed across the top of the output.
- 👉 **(`Vals`)** (Drop zone): numeric field(s) to summarise (when needed by the chosen aggregator).
- 👉 **(Drag within a zone)**: reorders fields.
- 👉 **(`×`)** (Button): removes a field from a zone.

Choose the calculation ("Aggregator")

- 👉 **(`Aggregator`)** (Dropdown): chooses how each cell is calculated. **Optional** — you can leave this at the default `Count` (no additional variable needed).
    - 👉 **(`Count`)**: how many rows fall into each cell.
    - 👉 **(`Sum` / `Average` / `Min` / `Max`)**: summarises a numeric field.
    - 👉 **(`Unique Count`)**: counts distinct values of a field.
- 👉 **(`Vals`)** (Dropdown): choose which numeric field to summarise (**optional**, only shown when needed).

Filter or exclude values

- 👉 **(Filter popup on a field chip)** (Popup): include/exclude values for that field.
- 👉 **(Search)** (Text field): narrows the value list.
- 👉 **(Checkbox list)**: include/exclude specific values (includes a "select all" control).

Sorting

- 👉 **(`Row Order`)** (Dropdown): sorts row keys (e.g. by key or by value, depending on the option).
- 👉 **(`Col Order`)** (Dropdown): sorts column keys.

Heatmaps and charts

- 👉 **(`Renderer`)** (Dropdown): switches between:
    - 👉 **Heatmaps** (e.g. `Heatmap`, `Row Heatmap`, `Col Heatmap`)
    - 👉 **Plotly charts** (e.g. `Bar`, `Line`, `Scatter`, `Stacked Bar`, `Area`, `Multiple Pie`)

Conditional formatting

One **Shading** dropdown in the toolbar colours the tables in all three subtabs the same way, so the pivot table, the Summary heatmaps and the All-by-all tables always match.

- 👉🏼 **(`Off`)**: no colouring.
- 👉🏼 **(`Rows`)** (default): shades each cell against the other cells in its own row, so you read across a row to compare.
- 👉🏼 **(`Columns`)**: shades each cell against the other cells in its column, so you read down instead.
- 👉🏼 **(`Cells`)**: shades how far each cell sits from the count you would expect if the two variables were unrelated. Blue means more cases than expected, red means fewer, and white means about as expected. Hover a cell to see its observed and expected counts. This is the view that shows you where an association actually comes from.

Row and column totals always keep their own blue shading. The pivot table's own **Heatmap** renderers colour their cells themselves, so Shading leaves those alone. The setting is saved to the URL, so a bookmark reopens with the same formatting.

Export and sharing

- 👉🏼 **(`Copy to Clipboard`)** (Button): copies the current pivot output as an image.
- 👉🏼 **(`Copy as Table`)** (Button): copies the pivot as TSV — pastes as a correctly aligned table in Excel or Google Sheets (**Table renderer only**).
- 👉🏼 **(`Download XLSX`)** (Button): exports the pivot table to `.xlsx` with merged cells expanded so columns align (**Table renderer only**).
- 👉🏼 **(URL state)**: the pivot configuration is saved to the URL automatically, so you can bookmark/share it. That includes which subtab you were on, the shading mode, and the field tick boxes on Summary and All-by-all, so a bookmark reopens on the same view.

Significance testing

When you build a **2-way count table** (one field in Rows, one in Cols, Count aggregator), the app runs a statistical test and shows the result below the table.

- **Sig level** (dropdown): choose your significance threshold (0.10, 0.05, 0.01, or 0.001). The test result shows whether the association is significant at that level.
- **Which test?** The app picks the right test for your data:
  - **Chi-squared** when both variables are nominal (e.g. categories with no natural order).
  - **Mantel** (linear-by-linear) when one or both variables are ordinal (e.g. Likert scales, age bands). This test is more sensitive to ordered trends.
- **All-by-all checks**: the third subtab, which runs as soon as you open it. The app tests every pair of categorical variables with 2–10 values, lists them by p-value, and shows tables for the significant

ones. Useful for exploratory analysis when you have several group or rating variables. The **Rows** and **Cols** tick boxes above the results narrow which pairs are shown; leave them empty for all of them.

Notes on the datasets

- **Links**: every causal link plus metadata; includes AI fields (e.g. confidence) and reserved columns like `original_cause`, `original_effect`.
- **Factors**: unique factors with frequency, source count, citations, and `original_label` (ALL underlying original labels for the displayed factor, concatenated with line breaks, derived from the current stage's links like `original_cause/effect`).
- **Sources**: document metadata and flattened custom fields (`custom_*`).

💡 Tip: For results that match other panels, use **After Analysis Pipeline**.